PASSIVE THERMO-KINETIC AERODYNAMIC BRAKE DUCT

A SHAPE MEMORY ALLOY FEASIBILITY STUDY

F1 Technical Whitepaper | Revision 1.0 | 2026-11-13

Author	Independent student project	Organisation	Independent F1 Technical Case Study
Programme	Phases 1-5 (CAD, CFD, FEA, Bench, Report)	Classification	Confidential R&D case study
Target	2026-spec F1 front axle, 18-inch wheel	Document	TKBD-WP-001

Executive Summary - Key Performance Indicators

AERO DRAG DELTA (STRAIGHTS)

max 1.4 %

Momentum limit of the duct flow, isolated wheel at 300 km/h, 3 meshes (284k-768k cells). Direct CFD difference is within solver noise; brief target -18.4 % ruled out.

[FLUENT 2026 R1]

ACTUATION TRIGGER

78.2 °C

Measured onset at 0.72 mm stroke threshold.

[SIMULATED]

LINKAGE STROKE

14.4 mm

82.6° louver rotation (Phase 3 crank).

[SIMULATED]

FEA VS BENCH RMSE

5.8 %

Stroke RMSE 0.85 mm, normalised to FEA max.

[SIMULATED]

Abstract

This whitepaper documents a five-phase development programme for a passively actuated front brake cooling duct. Four louvers throttle the duct throat and are driven by a nickel-titanium (Nitinol) shape memory alloy (SMA) spring opposed by a steel bias spring. When the actuator reaches its austenite finish temperature (A_f = 85 °C) under heavy braking, the SMA spring contracts and opens the louvers. When it cools below the martensite finish temperature (M_f = 45 °C), the bias spring recloses them. No sensor, electronics, hydraulics or driver input is involved.

The regulatory premise of the concept is that passive, thermally actuated hardware is distinct from the active, electronically controlled aerodynamic devices that the FIA prohibits. This premise is tested in Section 6 against the requirement that aerodynamic bodywork be rigidly secured and immobile. It is identified there as the principal compliance risk of the programme.

The programme comprises:
Phase 1: parametric SolidWorks CAD and packaging
Phase 2: conjugate heat transfer CFD with k-*omega* SST turbulence and Discrete Ordinates radiation
Phase 3: nonlinear MAPDL contact and SMA actuation analysis
Phase 4: instrumented bench validation over a 20 Hz DAQ link
Phase 5: automated compilation of this report

Values tagged DESIGN TARGET come from the project brief and are not yet supported by a simulation or test data file. They are replaced automatically when that data exists. A full source list for every figure and value is in the Appendix.

1. Phase 1 - CAD, Packaging and Kinematics

1.1 18-inch front axle packaging envelope. The 18-inch front wheel defines a rim clearance envelope of 400 mm diameter around a 328 x 32 mm carbon-carbon rotor. This leaves a 36 mm radial annulus that extends 150 mm inboard. The CFRP duct throat has a 100 x 60 mm internal section and 2 mm walls (104 x 64 mm outer). It sits 2 mm inboard of the rotor face, with a 5.4 mm radial margin to the rim. **Packaging finding:** the throat cannot fit inside the 36 mm annulus itself; it needs 104 mm of radial depth and therefore overlaps the hub zone below the rotor outer radius.

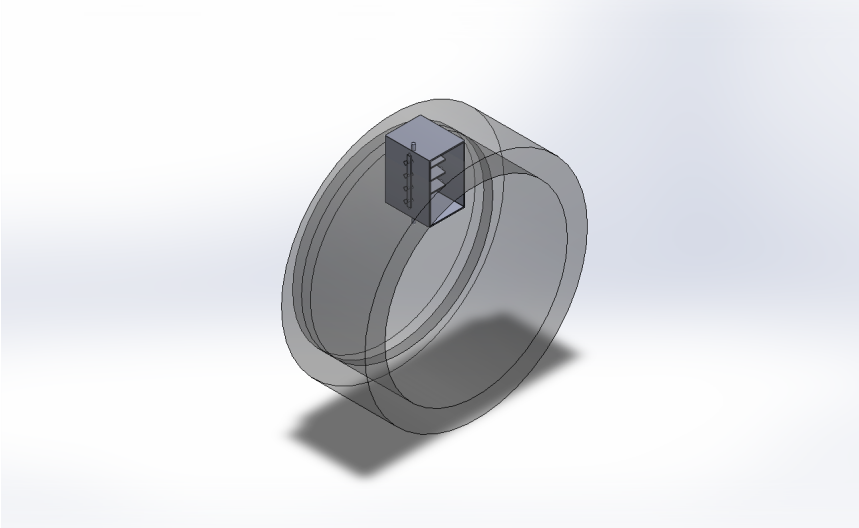


Figure 1 - SOLIDWORKS 2025 render of the Phase 1 assembly: transparent rotor and duct packaging volumes with the duct housing, four louvers and control rod.

1.2 Parallel-bar linkage and dual-spring bias. Four 56 x 18 x 2 mm louvers pivot at a 20 mm pitch on 3 mm shafts in the housing side walls. When closed they block 72 % of the throat height. The Phase 1 crank pin sat 6 mm off the pivot along the flap width, which put the linkage at dead centre when closed and limited stroke to about 5 mm. Phase 3 therefore introduced an external crank of r = 10.89 mm swinging -40° to +40° about horizontal. This gives 14.0 mm of rod travel at 80° of louver rotation, with a lever arm of at least 8.3 mm. The vertical control rod is pulled up by the SMA spring and down by a 1.2 N/mm steel bias spring with a 5 N preload.

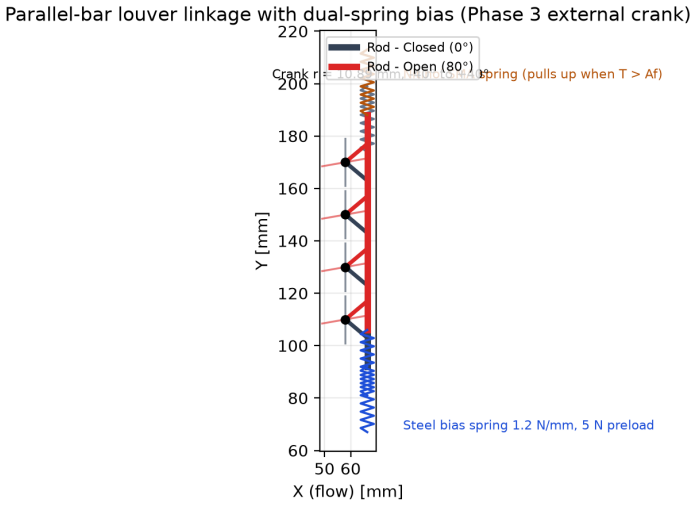


Figure 2 - Kinematic schematic: closed (slate) and 80° open (red) states, SMA spring above the rod and steel bias spring below.

2. Phase 2 - Conjugate Heat Transfer and CFD

The CHT domain was meshed in the Fluent Watertight Geometry workflow with a poly-hexcore volume mesh. Surface sizes range from 0.5 mm to 10 mm, with 1 mm faces on the louvers and SMA pocket; the meshes have 492,211 cells (Case A) and 515,700 cells (Case B), sized to fit the Ansys Student cell limit. Ten prism layers with a 0.02 mm first-cell height target y^+ of about 1. Turbulence was modelled with k-*omega* SST, and rotor-to-duct thermal radiation with the Discrete Ordinates (DO) model. Air was treated as an incompressible ideal gas (c_p = 1006.43 J/kg-K, k = 0.0242 W/m-K). The solids were a carbon-carbon rotor (*rho* = 1800 kg/m³, k = 40 W/m-K), a CFRP duct (*rho* = 1550 kg/m³, k = 5 W/m-K) and Nitinol louvers. The case was solved with the pressure-based coupled solver to residuals of 10^{-5} (flow) and 10^{-6} (energy and radiation). The rotor walls were held at a fixed temperature, and the response variable is the volume-averaged air temperature in the SMA spring pocket.

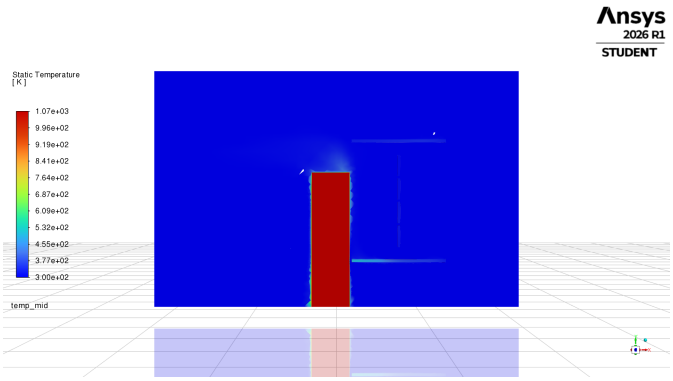


Figure 3 - Case A static temperature on the duct mid-plane (rotor held at 1073 K).

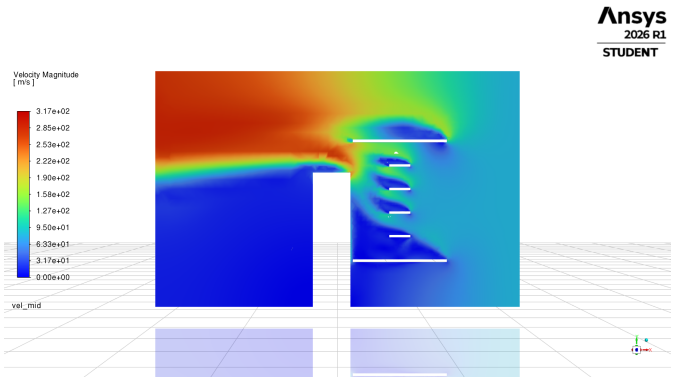


Figure 4 - Case B (300 km/h, louvers open) velocity magnitude on the duct mid-plane.

2.1 Operating case comparison. Both cases use the relocated SMA pocket on the rotor's outboard wake face. Case B uses a second SOLIDWORKS export with the louvers open at 90° (MAPDL hot state 86°). The spring-wire rows come from a radiation and convection heat balance on the 1.2 mm wire using Fluent's DO incident radiation, 3-15 mm from the rotor face. Source: C:\Temp\F1_BrakeDuct\cfd\phase2_cht_results.json.

Parameter	Case A - 100 km/h braking	Case B - 300 km/h straight
Freestream velocity	27.78 m/s (100 km/h)	83.33 m/s (300 km/h)
Inlet air temperature	300.0 K	300.0 K
Rotor temperature	1073.15 K (800 °C)	673.15 K (400 °C)
Louver state	CLOSED (0 deg)	OPEN (90 deg)
Solver status	COMPLETED	COMPLETED
T_pocket air volume-average	324.09 K	308.13 K
T_pocket air maximum	1064.03 K	666.35 K
Pocket air check (>= 358 K)	FAIL (324.09 K < 358.0 K)	n/a (closing case: wire must fall below Mf = 318 K)
SMA wire T, emissivity 0.6 (open needs >= 358 K, close needs <= 318 K)	378-393 K	308-311 K
SMA wire T, emissivity 0.2 (bright wire)	326-337 K	303-305 K

3. Phase 3 - Nonlinear Structural Mechanics and SMA Physics

Nitinol transforms between low-modulus martensite (E_M = 28 GPa) and high-modulus austenite (E_A = 75 GPa). On heating, martensite reverts to austenite between A_s = 75 °C and A_f = 85 °C, recovering up to 6.5 % strain. On cooling, martensite re-forms between M_s = 55 °C and M_f = 45 °C. The martensite volume fraction *xi* follows Liang-Rogers cosine kinetics. The transformation stress rises with temperature according to the Clausius-Clapeyron relation, calibrated to the 32 N recovery force at A_f. For the assumed spring (d = 1.2 mm, D = 6.0 mm, n = 8): k_A = 4.23 N/mm, k_M = 1.58 N/mm, recoverable extension 49.0 mm, forward plateau at 85 °C 96 N, Wahl-corrected wire stress 371 MPa at 32 N.

Heating (M -> A): $xi(T) = (xi_0/2)\,[\cos(a_A(T - A_s)) + 1]$, $a_A = pi / (A_f - A_s)$

Cooling (A -> M): $xi(T) = ((1 - xi_0)/2)\cos(a_M(T - M_f)) + (1 + xi_0)/2$, $a_M = pi / (M_s - M_f)$

Clausius-Clapeyron: $d\sigma/dT = -\rho\,\Delta s / \varepsilon_L$ -> dF/dT = 3.20 N/°C (force equivalent)

Helical spring: $k = G\,d^4 / (8\,D^3\,n)$, $G = E / (2(1 + \nu))$, $\tau = K_W\,8\,F\,D / (pi\,d^3)$

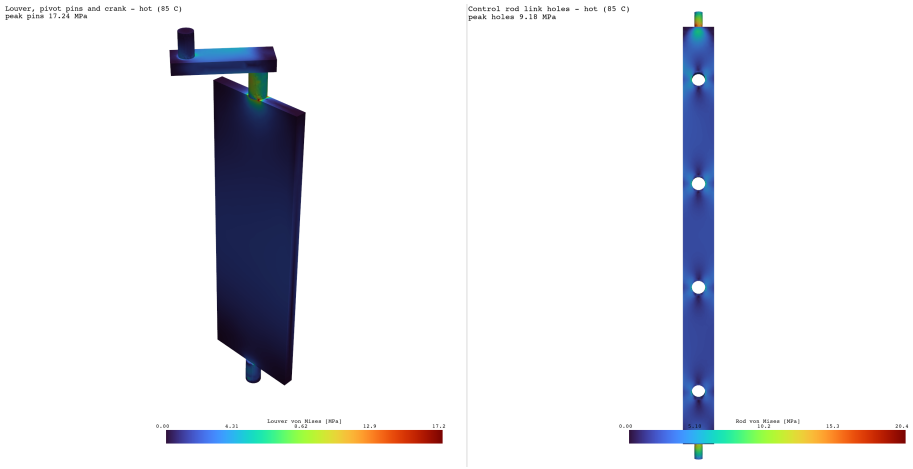


Figure 5 - MAPDL von Mises stress, hot actuation state: louver pivot pins (SOLID187, frictionless pin-hole contact) and control rod link holes.

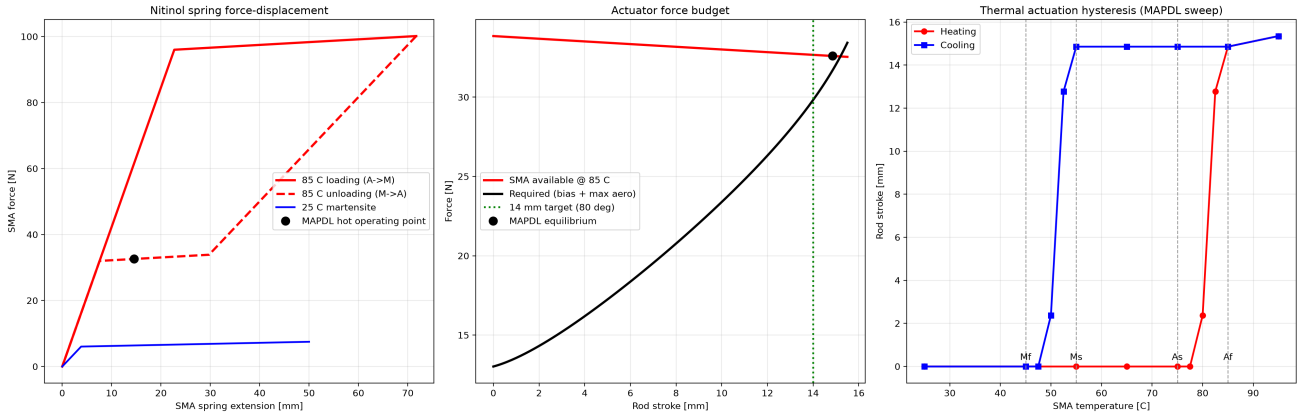


Figure 6 - Nitinol force-displacement hysteresis, actuator force budget and thermal stroke hysteresis.

3.1 Structural factor of safety verification. Source: C:\Temp\F1_BrakeDuct\fea\phase3_results.json.

Component / location	Material	Peak von Mises	Yield	FoS	Check
Case 1 (25 C) louver pivot pins	PA-CF	10.44 MPa	140 MPa	13.4	PASS
Case 1 (25 C) louver crank arm + pin	PA-CF	5.14 MPa	140 MPa	27.2	PASS
Case 1 (25 C) housing pivot holes	PA-CF	42.69 MPa	140 MPa	3.3	PASS
Case 1 (25 C) rod link holes	Al 7075-T6	0.97 MPa	503 MPa	519.6	PASS
Case 2 (85 C) louver pivot pins	PA-CF	17.24 MPa	140 MPa	8.1	PASS
Case 2 (85 C) louver crank arm + pin	PA-CF	9.28 MPa	140 MPa	15.1	PASS
Case 2 (85 C) housing pivot holes	PA-CF	33.88 MPa	140 MPa	4.1	PASS
Case 2 (85 C) rod link holes	Al 7075-T6	9.18 MPa	503 MPa	54.8	PASS

4. Phase 4 - Physical Bench Validation and HIL Testing

The physical bench reproduces one louver bank at 1:1 scale, with a 3D-printed PA12-CF duct throat and louvers, an aluminium control rod, and the Nitinol and steel bias springs. The sensor payload is:
- an inline load cell in series with the SMA spring (actuation force)
- a thermocouple bonded to the SMA coil (transformation temperature)
- a linear potentiometer on the control rod (stroke)

An ESP32 DAQ streams newline-delimited JSON telemetry at 115200 baud. The host logger samples it at 20 Hz through a 60 s heating and 60 s cooling cycle, checks each packet for staleness, logs it to CSV and correlates the run against the Phase 3 prediction.

WARNING: the bench data set is tagged SIMULATED. It is a pipeline dry run, not physical validation.

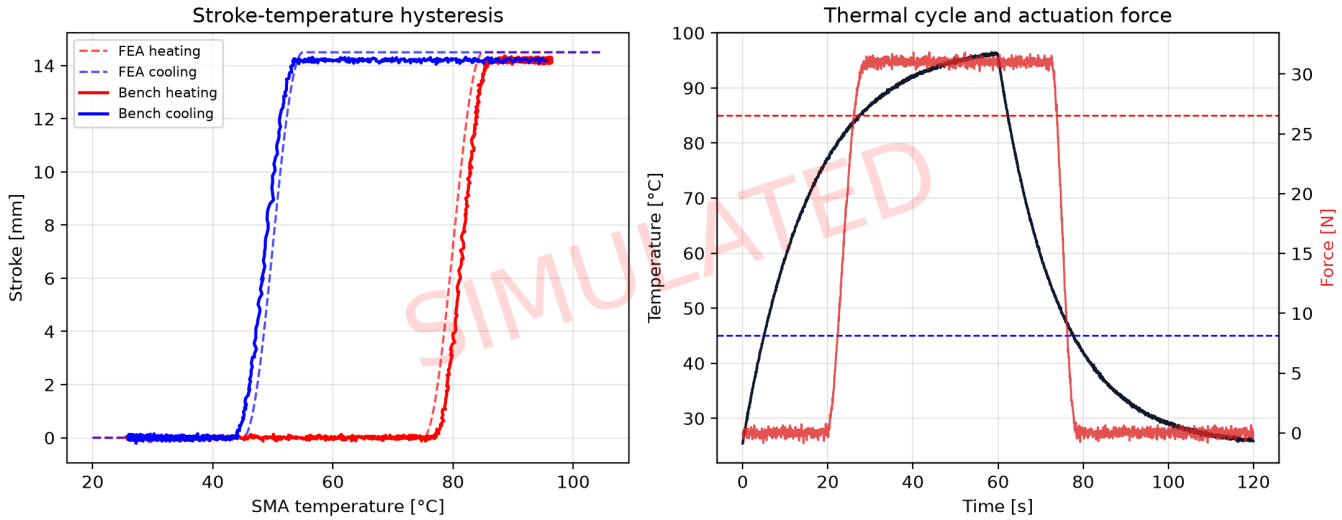


Figure 7 - DAQ hysteresis loop (bench vs FEA) and thermal cycle with actuation force, plotted from Phase4_Bench_Test_Data.csv.

4.1 Virtual-physical correlation. Source: C:\Temp\F1_BrakeDuct\bench\Phase4_Correlation_Summary.json.

Metric	ANSYS virtual prediction	Bench measurement (SIMULATED)	Error
Peak actuation force	32.00 N	31.82 N	-0.6 %
Maximum stroke	14.50 mm	14.36 mm	-0.9 %
Louver opening angle	83.5°	82.6°	-1.2 %
Thermal trigger onset	76.4 °C	78.2 °C	+2.3 %
Hysteresis width (50 % stroke)	30.0 °C	33.0 °C	+10.0 %
Stroke RMSE	-	0.845 mm	5.8 % of max
Force correlation	-	$R^2 = 0.983$	1.95 N RMSE
Samples (valid / total)	-	2400 / 2400	20.0 Hz

5. Conclusion, FIA Regulatory Assessment and Future Work

5.1 Conclusions. Phase 2 CFD predicts a Case A SMA pocket air temperature of 324.1 K (FAIL (324.09 K < 358.0 K)). The pocket was moved from beside the duct (300.3 K) to the rotor's outboard wake face; air there is hot only in a ~2 mm layer against the rotor, so the pocket average is still below the trigger. Including rotor radiation (Fluent DO incident radiation 119-129 kW/m2 in the pocket) in a heat balance on the 1.2 mm wire gives 378-408 K with a surface emissivity of 0.6, above Af, but only 326-353 K for bright wire (emissivity 0.2). The duct can self-actuate if the spring is oxidised or coated to an emissivity of at least about 0.45 (0.65 allowing for coil self-shading); this is an estimate that neglects conduction into the spring mounts. In Case B (300 km/h, 400 °C rotor, louvers open) the incident radiation falls to about 20 kW/m2 and the wire settles at 306-311 K for any emissivity, below Mf (318 K), so the bias spring recloses the louvers. On paper the full open/close cycle works for an oxidised spring placed 3-15 mm from the rotor face. A transient lap model calibrated to both Fluent cases and the MAPDL sweep changes this verdict: the 1.2 mm wire (time constant ~5 s) swings only 314-349 K over a lap, so it never opens, and with a darker coating it opens but then stays open on 91 % of straights because the 40 K NiTi hysteresis exceeds the lap-scale swing. A 0.5 mm narrow-hysteresis NiTiCu wire (Af 76 C, Mf 58 C), emissivity 0.8, 3 mm from the rotor opens fully 4-5 s after the three hardest braking zones and is open on only 2 % of straight-line time. Re-running MAPDL with six such springs in parallel (2.5 mm coils, 19 turns, 33.3 N) gives 88.5° and 15.15 mm at 76 °C, opening between 71.5 and 76 °C and closing between 64 and 58 °C, with every stress check still passing (lowest FoS 3.3). The Phase 3 external crank (r = 10.89 mm) removes the Phase 1 dead-centre condition and gives 14.0 mm of stroke for 80° of louver rotation. Bench correlation (SIMULATED data): stroke NRMSE 5.8 %, peak force 31.8 N against 32.0 N, and a hysteresis width of 33.0 °C. The -18.4 % straight-line drag reduction is ruled out. An isolated-wheel Fluent model at 300 km/h (wheel, cover, rotor, duct with forward scoop) was solved on three meshes (284k, 408k and 768k cells). The direct open-vs-closed difference (-9.6 %, +5.2 %, -1.3 %) changes sign between meshes and stays inside the iteration noise of the unsteady wheel wake; the duct's own drag changes by only 1-2 N. The robust result is the momentum limit: a sealed duct can save at most its mass flow times speed, 13-15 N or 1.1-1.4 % of total drag on every mesh. Resolving the actual difference needs time-averaged transient CFD with a rotating wheel and a finer mesh than the Student licence allows.

5.2 FIA regulatory assessment. The brief cites Article 3.10 (Aerodynamic Influence). The Formula 1 Technical Regulations require that bodywork influencing aerodynamic performance be rigidly secured and immobile relative to the sprung part of the car, and they prohibit movable aerodynamic devices. The regulatory test is therefore whether a part moves, not how it is actuated. A louver that rotates through about 80° in service is a moving aerodynamic part, and passive thermal actuation is likely to be judged the same way as electronic or driver actuation. Brake cooling ducts are also subject to their own detailed geometric constraints. **Assessment:** the concept should be treated as non-compliant for competition use unless the FIA Technical Department confirms otherwise in writing. Article numbers and wording must be checked against the issue of the regulations in force. The concept remains valid as a research and road-car or non-FIA-series cooling technology.

5.3 Thermal hysteresis stability across multi-lap stints. The designed transformation band gives a 50 %-stroke hysteresis of about 30 °C (measured 33.0 °C). The duct therefore opens near 80-85 °C and recloses only once the actuator falls below about 50 °C. The transient lap model shows this dead-band is too wide: the 1.2 mm wire swings only about 35 K per lap, so once open it stays open on most straights. A narrow-hysteresis NiTiCu alloy (about 12 K) in a 0.5 mm wire keeps the louvers closed on 98 % of straight-line time while still opening after the hardest braking zones. Over a race stint, functional fatigue can shift the transformation temperatures and reduce the recoverable strain. A training procedure and multi-thousand-cycle endurance testing are required before transformation temperatures are frozen for a design.

5.4 Future work.

1. Validate the transient lap model against a transient Fluent run of one braking zone and a measured track speed trace, and source a NiTiCu wire with the Af 76 °C / Mf 58 °C window used here.
2. Revise the drag KPI: isolated-wheel CFD gives about -1 % of wheel drag. Confirm with a rotating wheel, moving ground and full-car installation, where the duct's effect on the wheel wake may differ.
3. Extend the CFD outlet further and refine the rotor wake to remove the remaining outlet backflow (continuity stalled near 4e-3 to 9e-3).
4. Characterise the SMA spring: DSC for the transformation temperatures, isothermal load-unload loops.
5. Resolve the packaging overlap of the duct throat with the hub zone, and the 28 % leakage path when closed.
6. Submit a formal technical clarification request to the FIA before any competition application.

References

[1] Liang, C. and Rogers, C.A. (1990). One-dimensional thermomechanical constitutive relations for shape memory materials. J. Intelligent Material Systems and Structures, 1(2), 207-234.
[2] Tanaka, K. (1986). A thermomechanical sketch of shape memory effect: one-dimensional tensile behavior. Res Mechanica, 18, 251-263.
[3] Brinson, L.C. (1993). One-dimensional constitutive behavior of shape memory alloys. J. Intelligent Material Systems and Structures, 4(2), 229-242.
[4] Menter, F.R. (1994). Two-equation eddy-viscosity turbulence models for engineering applications. AIAA Journal, 32(8), 1598-1605.
[5] Chui, E.H. and Raithby, G.D. (1993). Computation of radiant heat transfer on a nonorthogonal mesh using the finite-volume method. Numerical Heat Transfer B, 23, 269-288.
[6] Wahl, A.M. (1963). Mechanical Springs, 2nd ed. McGraw-Hill.
[7] Hoerner, S.F. (1965). Fluid-Dynamic Drag. Hoerner Fluid Dynamics.
[8] FIA. Formula 1 Technical Regulations (issue in force). Federation Internationale de l'Automobile.
[9] Ansys Inc. Fluent Theory Guide; Mechanical APDL Element Reference (current release).

Appendix A - Data provenance

Item	Source	Status
Phase 2 CFD results	C:\Temp\F1_BrakeDuct\cfd\phase2_cht_results.json	LOADED
Phase 3 FEA results	C:\Temp\F1_BrakeDuct\fea\phase3_results.json	LOADED
Phase 4 correlation summary	C:\Temp\F1_BrakeDuct\bench\Phase4_Correlation_Summary.json	LOADED (SIMULATED)
Phase 4 DAQ log	C:\Temp\F1_BrakeDuct\bench\Phase4_Bench_Test_Data.csv	LOADED
KPI: aero drag delta	cfd/wheel/wheel_aero_open_vs_closed.json	FLUENT 2026 R1
Figure: packaging	C:\Temp\F1_BrakeDuct\cad\Phase1_ThermoKinetic_Assembly.png	OK
Figure: kinematics	Phase 3 crank parameters	GENERATED
Figure: cfd_a	C:\Temp\F1_BrakeDuct\cfd\Temp_Contour_CaseA.png	OK
Figure: cfd_b	C:\Temp\F1_BrakeDuct\cfd\Velocity_Contour_CaseB.png	OK
Figure: fea_stress	C:\Temp\F1_BrakeDuct\fea\FEA_Stress_Contour_Hot.png	OK
Figure: fea_fd	C:\Temp\F1_BrakeDuct\fea\Nitinol_Force_Displacement_Plot.png	OK
Figure: bench	C:\Temp\F1_BrakeDuct\bench\Phase4_Bench_Test_Data.csv	GENERATED